Supplementary Material: Stage-Aware Early Warning of Australian Winter Crop Yield Shortfall

1. Fixed-Model and Feature-Regime Details

Table S1: Fixed-model lead-time comparison. NYH denotes the no-yield-history weather-soil regime; Op. denotes the operational regime. Values are test RMSE.

Window	Ridge NYH	Ridge Op.	LGBM NYH	LGBM Op.
May-Jun	1.01	0.74	0.92	0.86
May-Jul	1.00	0.72	1.06	0.85
May-Aug	1.01	0.75	0.98	0.89
May-Sep	0.96	0.70	0.94	0.89
May-Oct	0.89	0.66	0.91	0.83

Table S2: Feature groups used for stage-aware early-warning models.

Feature group	Examples	Interpretation
Rainfall/dryness	rain_sum, rain_days, dry_days, dry-spell length	Water supply and dry-spell stress
Heat/cold	tmax_mean, heat_days_30, heat degree days, frost days	Heat exposure and frost/cold risk
Energy/evaporation	radiation_sum, evap_sum, vapor pressure	Growth energy and evaporative demand
Compound stress	hot-dry days, high-evaporation dry days	Combined weather stress
Soil background	AWC, clay/sand, SOC, pH, bulk density, depth	Regional vulnerability conditioning

Table S3: Feature-regime definitions used for ablation and information-source analysis.

Feature set	Weather	Soil	Yield hist.	Interpretation
Identity/time only	No	No	No	Crop-region persistence benchmark
Weather	Yes	No	No	Stage-weather signal only
Weather+Dev	Yes	No	No	Adds train-derived anomaly features
Weather+Soil	Yes	Yes	No	Adds static regional vulnerability
No-yield-history weather-soil	Yes	Yes	No	Main information-isolation regime
Yield history	No	No	Yes	Persistent productivity memory
Operational with yield history	Yes	Yes	Yes	Main deployment-oriented regime

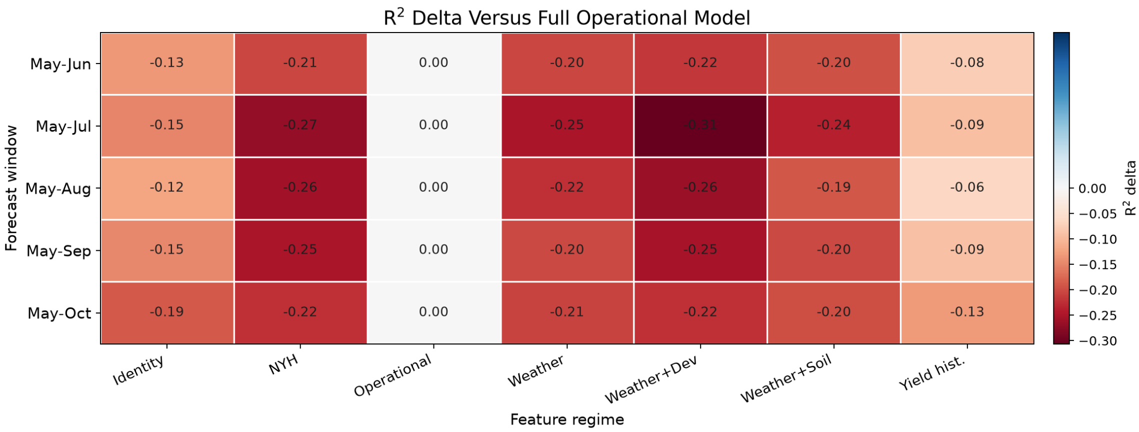

Figure S1: Ablation heatmap showing R^2 differences relative to the full operational feature set. Short labels are used to keep the dense comparison readable.

2. Data Coverage and Baseline Suite

Table S4: Missing source yield entries by state-crop series. The total confirms 966 observed entries out of 990 possible state-crop-year combinations in 1989–2021 for the observed crop-region pairs.

Region	Crop	Observed years	Missing years
New South Wales	Barley	33	0
New South Wales	Canola	33	0
New South Wales	Lupins	33	0
New South Wales	Oats	33	0
New South Wales	Wheat	33	0
Queensland	Barley	33	0
Queensland	Canola	30	3
Queensland	Lupins	24	9
Queensland	Oats	33	0
Queensland	Wheat	33	0
South Australia	Barley	33	0
South Australia	Canola	33	0
South Australia	Lupins	33	0
South Australia	Oats	33	0
South Australia	Wheat	33	0
Tasmania	Barley	33	0
Tasmania	Canola	33	0
Tasmania	Lupins	21	12
Tasmania	Oats	33	0
Tasmania	Wheat	33	0
Victoria	Barley	33	0
Victoria	Canola	33	0
Victoria	Lupins	33	0
Victoria	Oats	33	0
Victoria	Wheat	33	0
Western Australia	Barley	33	0
Western Australia	Canola	33	0
Western Australia	Lupins	33	0
Western Australia	Oats	33	0
Western Australia	Wheat	33	0
Total	All crops	966	24

Table S5: Compact internal baseline-suite summary by forecast window and feature regime. Full metrics are retained in the accompanying CSV artifacts.

Group	Regime	Window	Model	MAE	RMSE	R^2
Sequence	GRU no-yield	May-Jun	Weather-GRU	0.78	1.14	0.19
Sequence	GRU no-yield	May-Jul	Weather-GRU	0.80	1.15	0.17

Group	Regime	Window	Model	MAE	RMSE	R^2
Sequence	GRU no-yield	May-Aug	Weather-GRU	0.69	1.04	0.32
Sequence	GRU no-yield	May-Sep	Weather-GRU	0.63	1.00	0.38
Sequence	GRU no-yield	May-Oct	Weather-GRU	0.67	1.02	0.35
Sequence	GRU operational	May-Jun	Weather-GRU	0.59	0.78	0.62
Sequence	GRU operational	May-Jul	Weather-GRU	0.53	0.72	0.68
Sequence	GRU operational	May-Aug	Weather-GRU	0.63	0.85	0.55
Sequence	GRU operational	May-Sep	Weather-GRU	0.49	0.72	0.68
Sequence	GRU operational	May-Oct	Weather-GRU	0.52	0.72	0.68
Tabular ML	NYH weather-soil	May-Jun	XGBoost	0.65	0.91	0.48
Tabular ML	NYH weather-soil	May-Jul	CatBoost	0.66	0.95	0.44
Tabular ML	NYH weather-soil	May-Aug	HistGB	0.61	0.97	0.42
Classical ML	NYH weather-soil	May-Sep	SVR-RBF	0.61	0.88	0.52
Classical ML	NYH weather-soil	May-Oct	SVR-RBF	0.55	0.82	0.58
Classical ML	Operational	May-Jun	ENet	0.53	0.72	0.68
Classical ML	Operational	May-Jul	Ridge	0.53	0.72	0.68
Classical ML	Operational	May-Aug	ENet	0.51	0.72	0.67
Classical ML	Operational	May-Sep	Ridge	0.47	0.70	0.69
Classical ML	Operational	May-Oct	Ridge	0.43	0.66	0.73
Historical	Past-only yield	May-Jun	Rolling3	0.61	0.78	0.62
Historical	Past-only yield	May-Jul	Rolling3	0.61	0.78	0.62
Historical	Past-only yield	May-Aug	Rolling3	0.61	0.78	0.62
Historical	Past-only yield	May-Sep	Rolling3	0.61	0.78	0.62
Historical	Past-only yield	May-Oct	Rolling3	0.61	0.78	0.62

Full baseline-suite metrics remain available in the accompanying CSV artifacts; the PDF keeps the compact summary to preserve readability.

3. Data and Implementation Details

The raw yield source is the ABARES Australian Crop Report state-crop-data workbook `03_AustCropRpt20260303_StateCropData_v1.0.0.xlsx`, accessed on 4 July 2026; the modeled panel is the derived `yield_panel.csv`. Fields used are crop area, production, and yield in tonnes per hectare, filtered to the five winter crops and six state-level regions in the main paper. The harvest year is represented by `year_start`. Production and area are retained for audit but excluded from model features.

SILO/LongPaddock weather inputs are daily rainfall, maximum and minimum temperature, radiation, evaporation, and vapor pressure. They are aggregated from May through each forecast cutoff. The benchmark uses state-level regional daily series as a monitoring exposure proxy; latitude and longitude metadata are retained, but no crop-area-weighted gridded exposure is claimed. Weather-deviation features use training years only: for feature value $x_{r,w,t}$, the deviation is $x_{r,w,t} - \bar{x}^{train}_{r,w}$ for the same region and forecast window. Soil/ASRIS attributes are static regional background covariates from the Soil and Landscape Grid of Australia, including available water capacity, texture, bulk density, soil organic carbon, nutrients, pH, cation exchange capacity, and depth variables. These features condition regional vulnerability and may partly proxy region identity.

The no-yield-history regime has 73 predictors before one-hot expansion: crop, state-level region, year, 30 stage-weather summaries, 18 train-derived weather-deviation features, and 22 soil background attributes after excluding coordinate metadata. The operational regime adds one-year lag, three-year rolling past mean, and expanding past mean, for 76 predictors before one-hot expansion. Scaling, imputation, encoders, baselines, thresholds, and conformal residuals are fitted without test years; one-hot encoders handle unknown categories in stress tests. Configured weather thresholds are rain day 1 mm, heavy rain 10/25 mm, heat 25/30/35 C, frost 0 C, cold 5 C, and high evaporation 5 mm.

4. Baseline-Suite Implementation

The internal baseline suite is run once with random seed 42. This single-seed design is a limitation; model selection is by validation RMSE within each forecast window and feature regime, and test metrics are reported once. Tabular preprocessing uses median imputation and standard scaling for numeric predictors, one-hot encoding for crop and region, and unknown-category handling in held-out stress tests. The suite uses scikit-learn 1.9.0, XGBoost 3.3.0, LightGBM 4.6.0, CatBoost 1.2.10, pyGAM 0.12.0, and PyTorch 2.12.1+cpu in the recorded run.

Candidate models include Ridge with alpha 1.0 or 3.0; ElasticNet with alpha 0.01/l1 ratio 0.2 or alpha 0.03/l1 ratio 0.4; RandomForest with 400 trees and either unrestricted depth/min leaf 3 or depth 8/min leaf 5; SVR-RBF with C 1 or 10 and epsilon 0.1; HistGradientBoosting; XGBoost; LightGBM; CatBoost; and a pyGAM additive model over transformed features. The GRU sequence comparator uses daily weather up to the cutoff window, static covariates concatenated after the sequence encoder, and for the operational regime adds lag/rolling/expanding yield-history features to the static block. Two GRU configurations are validation-selected: hidden size 16 or 32, dropout 0.15 or 0.20, Adam learning rate 0.001, weight decay 1e-4, up to 160 epochs with patience 16.

5. Classification and Uncertainty Details

Table S6: Sensitivity of the low-yield-risk label to crop-specific training shortfall percentiles. The 80th percentile is the main setting.

Percentile	Train rate (%)	Validation rate (%)	Test rate (%)
70	30.1	13.8	24.2
80	20.3	11.2	18.1
90	10.4	2.6	9.4

Table S7: Best threshold-tuned classification metrics by forecast window. Pred.+ is the fraction of test examples flagged positive by the selected threshold.

Window	Model	Strategy	Thr.	Prec.	Recall	F1	ROC-AUC	PR-AUC	Brier	Pred.+
May-Jun	CatBoost	best F1	0.14	0.21	0.89	0.33	0.51	0.19	0.149	0.79
May-Jul	LightGBM	best F1	0.30	0.27	0.70	0.39	0.63	0.25	0.144	0.47
May-Aug	LogisticRegression	precision ≥ 0.3	0.22	0.27	0.78	0.40	0.65	0.25	0.147	0.52
May-Sep	HistGradientBoosting	best F1	0.63	0.28	0.74	0.41	0.69	0.27	0.197	0.48
May-Oct	LogisticRegression	precision ≥ 0.3	0.27	0.47	0.56	0.51	0.80	0.39	0.135	0.21

Table S8: Low-yield-risk confusion matrix using validation-selected thresholds.

Window	Model	Thr.	TN	FP	FN	TP	Prec.	Recall	F1
May-Oct	LogisticRegression	0.27	105	17	12	15	0.469	0.556	0.508

Table S9: Held-out May-Oct ranked watch-list example. Rank scores are uncalibrated classifier scores used for ordering analyst-review lists and should not be interpreted as fully calibrated event probabilities. Observed indicates the held-out low-yield-risk label.

Rank	Region	Crop	Year	Rank score	Alert	Observed	Shortfall
1	VIC	Oats	2018	1.000	1	1	0.465
2	NSW	Lupins	2018	1.000	1	1	0.410
3	VIC	Lupins	2018	1.000	1	0	0.274
4	VIC	Canola	2018	1.000	1	0	0.012
5	NSW	Wheat	2018	0.273	1	1	0.979
6	QLD	Wheat	2019	0.273	1	1	0.957
7	QLD	Wheat	2018	0.273	1	1	0.880
8	QLD	Barley	2019	0.273	1	1	0.865
9	TAS	Oats	2019	0.273	1	1	0.730
10	QLD	Canola	2018	0.273	1	1	0.646

Coverage below the nominal 80% target, especially in later windows, is interpreted as temporal calibration difficulty under small validation samples. These intervals are diagnostics for analyst review rather than reliable calibrated intervals for automatic decisions.

Table S10: Compact conformal uncertainty diagnostics by forecast window. Coverage and interval width are evaluated on held-out test years after validation-period calibration.

Window	Interval model	Nominal	Coverage	Width	qhat	Brier
May-Jun	Conformal ElasticNet	0.800	0.799	1.565	0.782	0.149
May-Jul	Conformal ElasticNet	0.800	0.792	1.423	0.711	0.144
May-Aug	Conformal ElasticNet	0.800	0.711	1.259	0.629	0.147
May-Sep	Conformal ElasticNet	0.800	0.758	1.251	0.626	0.197
May-Oct	Conformal ElasticNet	0.800	0.752	1.125	0.563	0.135

6. Supplementary Figures

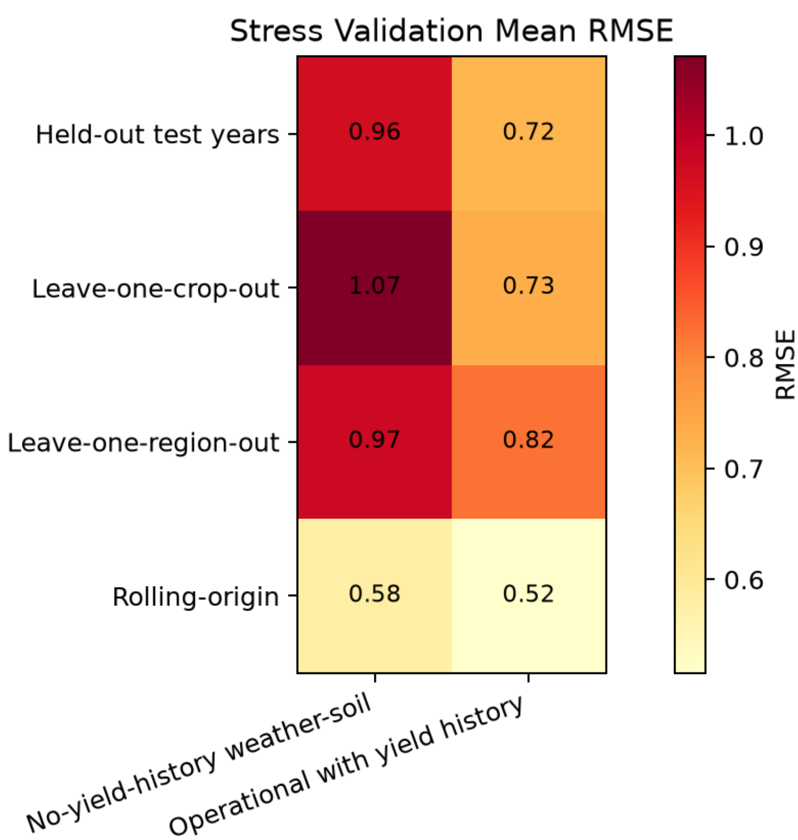

Figure S2: Stress validation heatmap by validation protocol and feature regime. Values use the same best-model, all-window summary as the main-paper stress validation table, not single-window May-Oct scores.

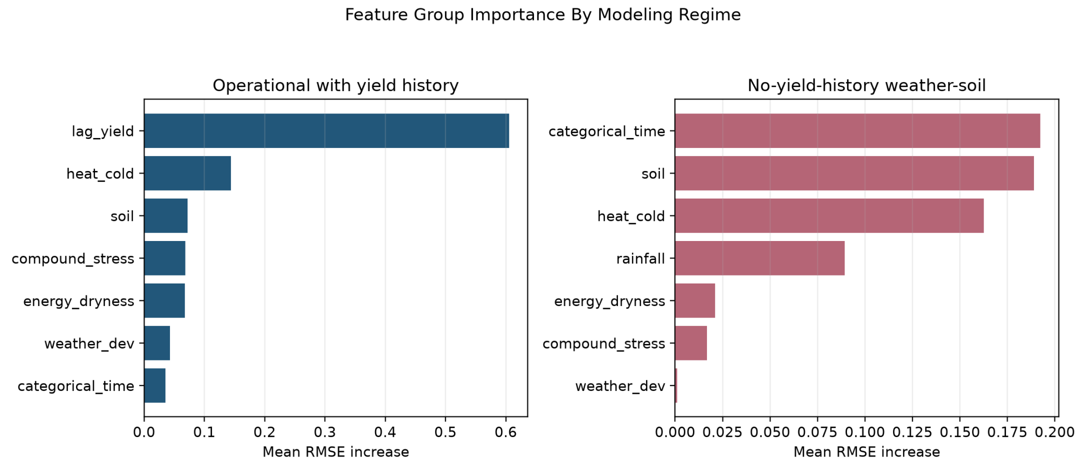

Figure S3: Feature group importance measured as mean RMSE increase after group permutation. Operational importance should not be interpreted as within-season weather importance because lag-yield features are allowed in that regime.